Individual Assessment Coversheet

To be attached to the front of the assessment.

Campus: Tyger Valley

Faculty: Information Technology

Module Code: ITPLA1-22

Group: 4

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Indicate	Yes	No
Plagiarism report attached		

Declaration:

I declare that this assessment is my own original work except for source material explicitly acknowledged. I also declare that this assessment or any other of my original work related to it has not been previously, or is not being simultaneously, submitted for this or any other course. I am aware of the AI policy and acknowledge that I have not used any AI technology to generate or manipulate data, other than as permitted by the assessment instructions. I also declare that I am aware of the Institution's policy and regulations on honesty in academic work as set out in the Conditions of Enrolment, and of the disciplinary guidelines applicable to breaches of such policy and regulations.

Signature	Date
Batista	22 June 2024

Lecturer's Comments:

Marks Awarded:	%

Signature	Date

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Section A

Question 1

1.1)

```
DetermineHighestAge ~Determine the highest age and index of the higher age in an array.
Begin

        Integer HighestAge, Index, childrenAges(13, 12, 15, 18, 15, 11, 19, 12, 17, 14, 10, 18)

        HighestAge = 0
        for i = 0 to 11 ~ Loop through all values of the childrenAges array
                if HighestAge < childrenAges(i) ~If false, update the Index and HighestAge values
                        Index = i
                        HighestAge = childrenAges(i)
                Endif
        Nexti

        Display “The highest age is ”, HighestAge
        Display “The index of the highest age is ”, Index

End (The University of Texas at Austin, 2017)
```

1.2)

```
CaclulatePayments ~Determine the totals for all payments that need to be made
Begin
        ~Declaration of variables
        Real Salary, Expenses, Savings, Emergency, Investments, Vacations

        Display “Enter you monthly salary: ”
        Enter Salary
        Salary = Salary – 1200 ~The paying required for housing

        Expenses = Salary * 30/100
        Display “Payment towards general expenses is ”, Expenses
        Savings = Salary * 20/100
        Display “Payment towards savings is ”, Savings
        Emergency = Salary * 15/100
        Display “Payment towards the emergency fund is ”, Emergency
        Investments = Salary * 20/100
        Display “Payment towards investments is ”, Investments
        Vacations = Salary * 15/100
        Display “Payment towards vacations is ”, Vacations
End
```

1.3)

where A = 3, C = 7, D = - 8, E = - 6, G = TRUE

D < (E - 7) OR G AND (A * C) >= (D – E * C)
-8 < (-6 - 7) OR TRUE AND (3 * 7) >= (-8 – (-6) * 7)
-8 < -13 OR TRUE AND (3 * 7) >= (-8 – (-6) * 7)
-8 < -13 OR TRUE AND 21 >= (-8 - (-6) * 7)
-8 < -13 OR TRUE AND 21 >= (-8 + 6 * 7)
-8 < -13 OR TRUE AND 21 >= (-8 + 42)
-8 < -13 OR TRUE AND 21 >= 34
FALSE OR TRUE AND 21 >= 34
FALSE OR TRUE AND FALSE
FALSE OR FALSE
FALSE

1.4)

where A = 5, B = 2, C = 12, D = -4, E = FALSE and F = TRUE

C * 4 >= B^3 – 18 \ 5 AND F OR A <= D – 5
12 * 4 >= 2^3 – 18 \ 5 AND TRUE OR 5 <= -4 – 5
48 >= 2^3 – 18 \ 5 AND TRUE OR 5 <= -4 – 5
48 >= 8 – 18 \ 5 AND TRUE OR 5 <= -4 – 5
48 >= 8 – 3 AND TRUE OR 5 <= -4 – 5
48 >= 8 – 3 AND TRUE OR 5 <= -9
48 >= 5 AND TRUE OR 5 <= -9
TRUE AND TRUE OR 5 <= -9
TRUE AND TRUE OR FALSE
TRUE OR FALSE
TRUE

Question 2

2.1)

```
CalculatePricePerLitre ~ Calculate the price in rand per litre of paint
Begin
        ~Declaration of variables
        Constant LitresPerCan = 5, Markup = 25/100
        Real PaintCanPrice, PaintCanPriceMarkup
        Integer LitrePrice

        Display "Enter the total cost per paint can: "
        Enter PaintCanPrice ~Input the price per paint can

        ~Calculates the price per paint can with the markup
        PaintCanPriceMarkup = (PaintCanPrice + (PaintCanPrice * Markup))

        ~Calculates price per litre of paint
        LitrePrice = (PaintCanPriceMarkup \ LitresPerCan) + 1

        Display "The price per litre of paint "LitrePrice ~ Output the price per litre of paint
End
```

2.2)

Instructions	**Expense**	**Increase by 25%**	**Gross price per litre**	**Price per litre of paint**	**Output**
Display					Enter the total cost per paint can:
Enter PaintCanPrice	182.60				
Calculate PaintCanPriceMarkup		228.25			
Calculate LitrePrice			45.65	46	
Display					The price per litre of paint 46

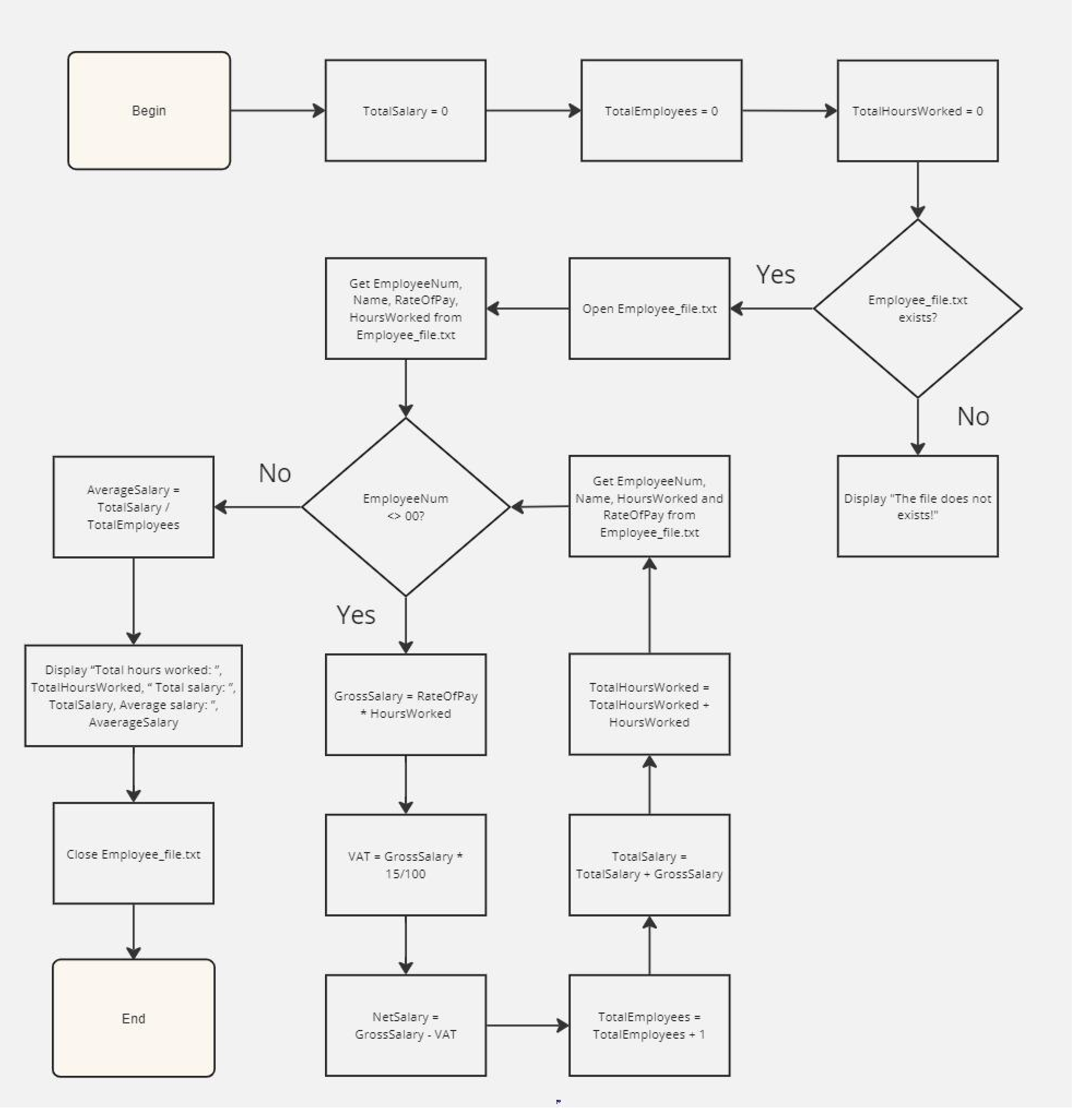
(101 Computing.net, 2019)

Question 3

3.1)

Flowchart was made in Miro

(CM Pretorius, 2023)



3.2)

```
CalculateHoursAndSalary ~Calculate and show the total hours worked, gross and average salary
Begin
	~Declaration of variables
	Real TotalSalary, AverageSalary, GrossSalary, NetSalary, VAT, RateOfPay
	Integer TotalHoursWorked, TotalEmployees, HoursWorked
	String EmployeeNum, Name

	~Initialise the variables
	TotalSalary = 0
	TotalEmployees = 0
	TotalHoursWorked = 0

	~ Determine if Employee_file.txt exists
	if exists(Employee_file.txt) then
		open Input(Employee_file.txt)

		~ Store all the values within the Employee_file.txt into variables
		call GetEmployeeFields(EmployeeNum, Name, HoursWorked, RateOfPay)

		do while EmployeeNum <> 00
			GrossSalary = HoursWorked * RateOfPay
			VAT = GrossSalary * 15/100
			NetSalary = GrossSalary – VAT
			TotalEmployees = TotalEmployees + 1
			~ Update the TotalSalary and TotalHoursWorked values
			TotalSalary = TotalSalary + GrossSalary
			TotalHoursWorked = TotalHoursWorked + HoursWorked
			call GetEmployeeFields(EmployeeNum, Name, HoursWorked, RateOfPay)
		loop

		~ Calculate the average salary
		AverageSalary = TotalSalary / TotalEmployees
		Display “Total hours worked: ”, TotalHoursWorked, “ Total salary: ”, TotalSalary,
		Average salary: ”, AverageSalary
		close(Employee_file.txt)
	Endif
	else
		Display “The file does not exist!”
	Endif
End (PseudoEditor.com, 2024)
```

3.3)

Instructions	EmployeeNum	Name	HoursWorked	RateOfPay	GrossSalary	Net Salary	VAT	TotalSalary	TotalEmployees	TotalHoursWorked	AverageSalary	If file exists	If EmployeeNum <> 00	Output
Initialise TotalSalary								0						
Initialise TotalEmployees									0					
Initialise HoursWorked										0				
Determine if Employee_file.txt exist												True		
Get the values associated with an employee	101	Johan	184	20										
Determine if EmployeeNum <> 00													True	
Calculate GrossSalary					3680									
Calculate VAT							552							
Calculate NetSalary						3128								
Calculate TotalEmployees									1					
Calculate TotalSalary								3680						
Calculate TotalHoursWorked										184				
Get the values associated with an employee	102	Sithole	200	25										
Determine if EmployeeNum <> 00													True	
Calculate GrossSalary					5000									
Calculate VAT							750							
Calculate NetSalary						4250								
Calculate TotalEmployees									2					
Calculate TotalSalary								8680						
Calculate TotalHoursWorked										384				
Get the values associated with an employee	103	Kamogelo	222	32										
Determine if EmployeeNum <> 00													True	
Calculate GrossSalary					7104									
Calculate VAT							1065.6							
Calculate NetSalary						6038,4								

Calculate TotalEmployees									3					
Calculate TotalSalary								15784						
Calculate TotalHoursWorked										606				
Get the values associated with an employee	00													
Determine if EmployeeNum <> 00													False	
Calculate AverageSalary											5261,33			
Display the TotalHoursWorked, TotalSalary and AverageSalary														"Total hours worked: ", 606, " Total salary: ", 15784, " Average salary: ", 5261,33

Question 4

```
Function CalculateAverage(Total)
	Real Average = Total / 10 ~Calculate the average of the ten marks
returnAverage

Sub DetermineGrade(Average) ~Displays the grade based off of the value of the average

	if Average <= 100 AND Average >= 90
		Display “Distinction”
	Endif
	if Average <= 89 AND Average >= 80
		Display “Very Good”
	Endif
	if Average <= 79 AND Average >= 70
		Display “Good”
	Endif
	if Average <= 69 AND Average >= 60
		Display “Credit”
	Endif
	if Average <= 59 AND Average >= 50
		Display “Pass”
	Endif
	if Average <= 49
		Display “Fail”
	Endif

Endsub

MainModule ~Main program
FinalMark
Begin
	~Declaration of variables
	Integer TotalMarks, testmarks(10)
	Real AverageMarks,

	TotalMarks = 0
	for i = 0 to 9 ~Loops ten times to fill the testmarks array and calculate the total of all marks
		Display “Enter your test ”, i, “ mark: ”
		Enter testmarks(i)
		TotalMarks = TotalMarks + testmarks(i)
	Nexti

	AverageMarks = CalculateAverage(TotalMarks) ~Calculate the student’s average mark
	Call DetermineGrade(AverageMarks) ~Calls the sub-procedure to determine the grade
End (PseudoEditor.com, 2024)
```

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